

Hossam Salah

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PROFESSIONAL SUMMARY

Data Analyst with nearly 3 years turning messy multi-system operational data into executive decisions, using SQL, Snowflake, Python, and Power BI. Identified \$70K in unbilled revenue and \$30K in inefficient marketing spend, improved forecasting accuracy by 27%, and diagnosed an 11% drop in customer visit frequency that pivoted a multi-region growth strategy. Builds the full path from data modeling and ETL through to the dashboards leadership actually uses, in a regulated US healthcare environment.

PROFESSIONAL EXPERIENCE

Data Analyst, COB Solution (US Healthcare) | US time zone

Dec 2023 – Present

- **Identified \$70K in unbilled revenue** by reconciling service records against payment transactions across a 30+ location network, ranking every case by evidentiary certainty so collections staff worked the highest-confidence recoveries first.
- **Pivoted a multi-region growth strategy by diagnosing an 11% drop** in customer visit frequency; cohort analysis proved new locations beat historical benchmarks by 29%, redirecting spend from acquisition to retention and surfacing \$30K in wasted marketing budget.
- **Improved demand forecasting accuracy 27%** with automated Python time-series models over 10K+ weekly records.
- **Cut query times 40% and reclaimed 10+ hours/week** by redesigning the Snowflake schema, automating 5+ manual reporting workflows, and standardizing 900+ inconsistent Excel files into a single pipeline.
- **Cut no-show losses 10%** by deploying a model that scores each day's schedule, letting staff confirm at-risk bookings or double-book the slot to convert would-be empty appointments into billable visits. Calibrated XGBoost, refined over 6 iterations with leakage and drift monitoring.
- **Lifted staff utilization from 80–85% to 90–95%** by pinpointing the profit-maximizing visit volume per location, modeling labor, operating hours, and overhead against revenue to show where added volume would force hours the site could not fill, and where trimming visits protected margin.
- Present findings to executive leadership in a weekly review, translating analysis into decisions on growth strategy, staffing, and revenue recovery.

SKILLS

- **Querying & Programming:** SQL, Snowflake, Python (Pandas, NumPy), Advanced Excel, ETL/ELT
- **Business Intelligence:** Power BI, DAX, Semi-Additive Measures, Tableau, Matplotlib, Seaborn, Dashboard Design
- **Data Modeling:** Dimensional Modeling, Snapshot Fact Tables, Slowly Changing Data, Window Functions, CTEs, Snowflake Schema Design
- **Analytical Methods:** Cohort Analysis, Retention Analysis, Funnel Analysis, Customer Segmentation, Unit Economics, A/B Testing, Statistical Inference, Time Series Forecasting, KPI Modeling, Financial Modeling, Break-Even Analysis
- **Applied Machine Learning:** Scikit-learn, XGBoost, Feature Engineering, Model Calibration, Model Explainability (SHAP), Classification, Regression
- **Tools:** Git, GitHub, Jupyter, Google Cloud Storage, ClickUp

KEY PROJECTS *(Full Portfolio)*

Revenue Mix & Unit Economics Analysis

- Consolidated 171 fragmented customer segments into a standardized taxonomy over 113K+ transactions.
- Found an 11x spread in revenue per transaction (\$25 to \$280) across segments, and identified the mix shift driving a sustained decline in average revenue.
- Ranked locations by revenue-mix efficiency and sized the upside of shifting mix, driving an operating plan across referral targeting, contract renegotiation, and staffing allocation.
- Delivered an executive dashboard with location scorecards and a monthly tracking cadence.

Collections Analytics Platform

- Replaced manual front-desk balance checks with a Snowflake model classifying every transaction line into 15+ dispositions to drive a structured cleanup workflow.
- Made deletions and corrections auditable in a source with no change history, using snapshot fact modeling and synthetic tombstone records to measure cleanup progress over time.
- Fixed a semi-additive aggregation error where plain sums broke under grouped contexts, re-architecting the DAX measure layer around a snapshot-date-in-context pattern.
- Documented 10+ load-bearing business rules with stakeholders before build; shipped a two-tier dashboard splitting executive KPIs from an operational data-quality view.

Executive KPI Platform

- Replaced manual spreadsheet reporting with real-time operational, service, and scheduling KPIs for a 30+ location network.
- Built end to end from the Snowflake data layer through a Python API to an interactive dashboard, with composite location-health scoring and rule-based alerting.

PERSONAL PROJECTS

Customer Churn Prediction | *Telecom*

- ML pipeline on 7,000+ customer records: EDA, feature engineering, class-imbalance handling, and tuning across seven algorithms. Result: 86% F1-score, 0.86 ROC AUC. Python, XGBoost, Scikit-learn.

Automatic Leukemia Image Detection

- CNN classifying blood-smear cell images as leukemic or healthy, benchmarked against classical models. Result: 88% accuracy, 0.87 F1-score. Python, TensorFlow, Keras, OpenCV.

EDUCATION

Associate Data Scientist, *DataCamp* | *Remote*

May 2024 – Sep 2024

- Completed a 90-hour track covering data management, EDA, statistical modeling, machine learning, and data communication.

Bachelor of Biomedical Engineering, *Al-Shorouk Academy* | *Cairo, Egypt*

Oct 2018 – Jul 2023

- Grade: Very Good | Developed an IoT wearable device as a graduation project, earning an excellent score.

LANGUAGES

Arabic: **Native**

English: **Proficient**